GROUP 2 | DATA SCIENCE CAPSTONE PROJECT

Water Quality & Potability Prediction

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Problem Statement

Access to clean and safe drinking water is one of the most critical global public health challenges. Contaminated water sources are responsible for millions of preventable deaths each year, particularly in developing regions where water quality monitoring infrastructure is limited. This project aims to build a machine learning classification model that predicts whether water is potable (safe to drink) based on physicochemical properties such as pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. This is a **binary classification problem** where the target variable is Potability (1 = potable, 0 = not potable). The goal is to provide a faster, data-driven approach to water safety assessment that can support decision-making by governments, NGOs, and water treatment agencies.

Project Objectives

The specific objectives of this project are:

1. **Conduct a thorough exploratory data analysis (EDA)** to understand data distributions, feature relationships, missing data patterns, and class balance.
2. **Clean and Preprocess the Data** by handling missing values, outliers, and preparing the dataset for machine learning by ensuring data quality and consistency.
3. **Train and compare three distinct machine learning classifiers**: KNN, Decision Tree, and Random Forest.
4. **Evaluate all models** using a comprehensive set of metrics: accuracy, precision, recall, F1-score, confusion matrix, and AUC-ROC.
5. **Derive actionable insights and recommendations** from model results to inform real-world water safety management.

Data Description

Dataset Overview

The dataset used in this study is the Water Potability Dataset, publicly available on Kaggle. It contains physicochemical measurements for 3,276 water samples collected from various sources. Each sample is described by nine continuous numerical features representing water quality parameters, and one binary integer target variable (Potability) indicating whether the sample is safe for drinking.

Property	Value
Total Samples (Rows)	3,276
Total Columns	10 (9 features + 1 target)
Feature Datatypes	Float64 (9 features), Int64 (1 target)
Missing Values Present	Yes — in 3 features
Target Variable	Potability (0 = Non-Potable, 1 = Potable)
Class 0 (Non-Potable) Count	1,998 samples (61.0%)
Class 1 (Potable) Count	1,278 samples (39.0%)

Feature-Level Observations

pH

pH measures the acidity or alkalinity of water on a scale of 0 to 14, where 7.0 is neutral. The WHO recommends a pH range of 6.5 to 8.5 for safe drinking water. Water with a pH below 6.5 is considered acidic and can corrode pipes, leaching heavy metals. Water above 8.5 is alkaline and may have an unpleasant taste.

Hardness

Water hardness is caused by dissolved calcium and magnesium salts. High hardness reduces the effectiveness of soap and can cause scale deposits in pipes but is not generally harmful to health.

Solids (Total Dissolved Solids — TDS)

TDS is a measure of all inorganic and organic substances dissolved in water. High TDS can affect water taste and may indicate the presence of harmful dissolved substances.

Chloramines

Chloramines are disinfectants formed when chlorine is combined with ammonia, commonly used in water treatment to kill bacteria and viruses. The EPA sets the maximum safe level at 4 ppm. Prolonged exposure to high chloramine levels is associated with respiratory issues and kidney damage.

Sulfate

Sulfates are naturally occurring ions in soil and rock. While not acutely toxic, very high concentrations can have a laxative effect.

Conductivity

Electrical conductivity reflects the total concentration of dissolved ionic species in water. Higher conductivity generally indicates higher dissolved solids and ionic contamination.

Organic Carbon

Total Organic Carbon (TOC) measures the amount of organic matter in water. For treated drinking water, WHO recommends below 2 mg/L. The dataset mean of 14.28 mg/L is far above this standard, again suggesting these samples may represent raw or undertreated water rather than finished drinking water.

Trihalomethanes (THMs)

THMs are disinfection by-products formed when chlorine reacts with organic matter in water. They are associated with cancer risk at high levels.

Turbidity

Turbidity measures how cloudy or clear the water is, caused by suspended particles. Turbidity is a key safety indicator because high turbidity can shield pathogens from disinfection.

Methodology

The methodology follows a structured, end-to-end data science pipeline consisting of six phases: Exploratory Data Analysis, Missing Value Analysis, Data Cleaning & Outlier Treatment, Feature Engineering, Preprocessing, and Model Training & Evaluation. Each phase is described in detail below.

Exploratory Data Analysis (EDA)

EDA was conducted across six systematic steps to build a thorough understanding of the dataset before any modelling decisions were made.

Step 1 — Data Loading

The dataset was loaded from Google Drive into a Pandas Data Frame using pd.read_csv(). The first ten rows were inspected using .head(10) to confirm that all columns were loaded correctly, data types were appropriate, and values appeared plausible. The dataset was confirmed to have 3,276 rows and 10 columns.

Step 2 — Initial Data Inspection

The .info() method was used to examine column names, non-null counts, and data types. Key observations from this step:

- All 9 feature columns are continuous float64 values, and the target (Potability) is int64.
- No categorical variables are present, eliminating the need for label encoding.
- Three columns — ph (2,785 non-null), Sulfate (2,495 non-null), and Trihalomethanes (3,114 non-null) — contain missing values, all others are complete.

Step 3 — Summary Statistics

The .describe() method produced descriptive statistics for all numerical columns. This revealed:

- pH has the widest relative range (0.00 to 14.00), suggesting extreme values requiring outlier treatment.
- All features have positive, non-zero values, ruling out impossible measurements.
- The 25th, 50th, and 75th percentiles for most features suggest roughly symmetric, near-normal distributions — later confirmed by histograms.

Step 4 — Missing Values Analysis

The .isna().sum() method was used to identify and quantify missing values. A heatmap of the missing value matrix was produced using seaborn to visually confirm which rows and columns were affected.

Results:

Feature	Missing Count	% of Total Rows	Imputation Strategy
ph	491	15.0%	Median imputation (SimpleImputer)
Sulfate	781	23.8%	Median imputation (SimpleImputer)
Trihalomethanes	162	4.9%	Median imputation (SimpleImputer)
All other features	0	0.0%	No imputation required

Median imputation was chosen over mean imputation because it is robust to outliers — particularly important for pH and Sulfate, which showed heavy tails before capping.

Step 5 — Class Balance Analysis

The target variable (Potability) was examined using .value_counts():

- Class 0 (Non-Potable): 1,998 samples — 61.0% of the dataset
- Class 1 (Potable): 1,278 samples — 39.0% of the dataset

This confirms a moderate class imbalance. The majority class (non-potable) is approximately 1.56 times the size of the minority class (potable).

Step 6 — Data Visualization

Six categories of visualizations were produced to support EDA:

Count Plot

A seaborn count plot confirmed the class imbalance described above, with approximately 1,998 non-potable and 1,278 potable samples displayed as a bar chart.

Histograms of All Features

A 2×3 grid of histograms was produced for pH, Chloramines, Turbidity, Hardness, Solids, and Sulfate. Key observations:

- pH distribution is approximately bell-shaped, centered near 7.0.
- Chloramines follow a near-normal distribution centered around 7 ppm.
- Turbidity is approximately normally distributed around 4 NTU.
- Hardness is approximately normal, centered around 196 mg/L.
- Solids are slightly right-skewed, reflecting the high-variance TDS values.
- Sulfate shows a near-normal distribution centered around 333 mg/L.

Correlation Heatmap

A seaborn heatmap of the full 10×10 correlation matrix (Pearson r) revealed a critical finding: all feature-to-feature correlations and all feature-to-target correlations are extremely weak, with absolute values below 0.20. The largest observed correlation with Potability was Solids (r = 0.034), which is negligible.

This indicates that:

- No single feature is a reliable linear predictor of water potability.
- Multicollinearity is not a concern — features are largely independent of each other.
- Non-linear models are more likely to capture the complex interactions needed to discriminate between classes.

Boxplots — Pre- and Post-Outlier Treatment

A 3×3 grid of boxplots was produced for all nine features before and after outlier capping. Pre-treatment boxplots showed significant upper whisker extension in pH, Solids, and Sulfate. Post-treatment boxplots confirmed effective outlier control while preserving data row count.

Data Cleaning and Outlier Treatment

Outlier Capping Strategy

Rather than removing outlier rows (which would have reduced the already modest dataset of 3,276 rows), an IQR-based capping strategy was applied. For each feature, values below Q1 − (multiplier × IQR) or above Q3 + (multiplier × IQR) were clipped to the respective boundary. This preserves all rows while controlling extreme values.

NOTE	Zero rows were removed during outlier treatment. The cleaned dataset retains all 3,276 original rows, ensuring maximum training data for model development. This is a deliberate design choice — the dataset is not large enough to afford row deletion.

Missing Value Imputation

After outlier capping, missing values in pH (491), Sulfate (781), and Trihalomethanes (162) were imputed using sklearn's SimpleImputer with strategy='median'. Median was preferred over mean because the capped distributions still exhibited some asymmetry, and median is more robust to any residual skewness. After imputation, all 10 columns had zero missing values.

Feature Engineering

Three binary safety flag features were engineered from existing continuous measurements by applying official WHO and EPA safety thresholds.

New Feature	Derived From	Threshold Condition	Regulatory Source	Value if Safe	Value if Unsafe
ph_safe	ph	6.5 ≤ pH ≤ 8.5	WHO Guidelines for Drinking Water Quality	1	0
chloramine_safe	Chloramines	Chloramines ≤ 4 ppm	EPA Stage 2 Disinfectants Rule	1	0
turbidity_safe	Turbidity	Turbidity ≤ 5 NTU	WHO Turbidity Guideline	1	0

These binary features give the model direct, interpretable signals grounded in established science. For example, if pH is outside the 6.5–8.5 range, the water fails a key WHO criterion regardless of its absolute value. The engineered features complement the original continuous features and were included in the final feature set alongside them.

Feature Selection and Scaling

The final feature set used for modelling comprised all nine original physicochemical features plus the three engineered safety flags — a total of 12 input features. The target variable was Potability.

MinMaxScaler from sklearn.preprocessing was applied to all nine original numerical features, transforming them to the [0, 1] range. This is particularly important for KNN, which computes Euclidean distances and would be dominated by high-magnitude features (such as Solids, with values up to 61,000 ppm) without scaling. The three binary engineered features (already in {0, 1}) were not rescaled.

Train-Test Split

The dataset was partitioned into training and testing sets using sklearn's train_test_split with:

- Training set: 80% of samples — 2,620 rows
- Test set: 20% of samples — 656 rows
- random_state = 42 — ensures exact reproducibility of results

Split	Rows	Features (X)	Target (y)
Training Set (X_train / y_train)	2,620	9 features	Potability labels
Test Set (X_test / y_test)	656	9 features	Potability labels

Model Training

Three machine learning classifiers were selected to provide a broad comparison across model complexity and algorithmic approach — KNN, Decision Tree and Random Forest.

Model 1: K-Nearest Neighbors (KNN)

K-Nearest Neighbours classifies a new sample by identifying the K most similar training samples (neighbors) in feature space, measured by Euclidean distance, and assigns the majority class label among those neighbors.

Hyperparameter	Value	Justification
neighbors	5	Standard default; balances bias-variance tradeoff for moderate-sized datasets
metric	Euclidean	Standard L2 distance; appropriate for continuous, scaled features
weights	uniform	All neighbours weighted equally
leaf_size	50	Controls speed of BallTree/KDTree construction; larger = faster build

Model 2: Decision Tree Classifier

A Decision Tree recursively partitions the feature space by selecting the feature and threshold at each node that maximizes the reduction in impurity (measured by the Gini criterion). It produces a hierarchical tree structure of binary decisions that is fully interpretable — each prediction can be traced from root to leaf.

Hyperparameter	Value	Justification
max_depth	5	Limits tree depth to prevent overfitting; controls model complexity
criterion	gini	Gini impurity: computationally efficient and commonly used for classification
random_state	42	Ensures reproducibility across runs

Model 3: Random Forest Classifier

Random Forest is a model that trains a large number of Decision Trees on combined subsets of the training data, with a random subset of features considered at each split. Predictions are made by majority voting across all trees. This approach reduces variance compared to a single tree, making Random Forest highly robust to overfitting and noise.

Hyperparameter	Value	Justification
n_estimators	100	100 trees provide stable ensemble predictions; standard baseline
random_state	42	Reproducibility

n_jobs	-1	Uses all available CPU cores for parallel training
max_depth	None (default)	Trees grow to full depth; ensemble averaging controls overfitting
criterion	gini (default)	Gini impurity for node splitting

Results and Evaluation

Evaluation Framework

Each model was evaluated on the held-out test set (656 samples) using five complementary metrics:

Metric	Definition	Why It Matters Here
Accuracy	Correct predictions / Total predictions	Provides an overall performance snapshot; misleading with class imbalance
Precision (Class 1)	True Positives / (True Positives + False Positives)	How often the model is correct when it predicts potable — affects trust in positive predictions
Recall (Class 1)	True Positives / (True Positives + False Negatives)	How many actual potable samples are correctly identified — critical for public health safety
F1-Score (Class 1)	Harmonic mean of Precision and Recall	Balances precision and recall; best single metric for imbalanced binary classification
AUC-ROC	Area under the Receiver Operating Characteristic curve	Measures model's ability to distinguish between classes across all thresholds; threshold-independent

Overall Performance Comparison

Model	Accuracy	Prec. (0)	Rec. (0)	F1 (0)	Prec. (1)	Rec. (1)	F1 (1)	Macro F1	AUC-ROC
KNN	53.5%	0.61	0.71	0.66	0.33	0.25	0.28	0.47	0.482
Decision Tree	63.7%	0.66	0.88	0.75	0.53	0.23	0.32	0.54	0.605
Random Forest	69.4%	0.70	0.89	0.78	0.66	0.36	0.47	0.63	0.686

Random Forest is the clear winner across every metric. The progression from KNN → Decision Tree → Random Forest shows a consistent improvement in accuracy (+9.9pp and +5.7pp respectively), precision, F1-score, and AUC-ROC.

Detailed Model Analysis

K-Nearest Neighbours (KNN)

Accuracy: 53.5% | AUC-ROC: 0.482

KNN was the weakest of the three models, performing at barely above random chance (50%) accuracy. The AUC-ROC of 0.482 is actually below 0.5, indicating that the model's class probability scores are slightly worse than random — meaning the model's confidence ordering of predictions is marginally inverted.

Confusion Matrix Analysis

	Predicted Non-Potable (0)	Predicted Potable (1)
Actual Non-Potable (0)	291 (True Negatives)	121 (False Positives)
Actual Potable (1)	184 (False Negatives)	60 (True Positives)

Of 412 actual non-potable samples, KNN correctly identified 291 (70.6% recall). Of 244 actual potable samples, only 60 were correctly predicted (24.6% recall). The high False Negative rate (184) is particularly concerning — these are potable water samples that KNN wrongly classified as non-potable. Conversely, 121 non-potable samples were falsely labelled potable (False Positives), which in a real-world context would represent a serious public health failure.

Interpretation

KNN's poor performance can be attributed to samples close in Euclidean distance not actually sharing the same potability class. Additionally, KNN is sensitive to irrelevant features, which are abundant in this dataset given the near-zero correlations observed during EDA.

Decision Tree Classifier

Accuracy: 63.7% | AUC-ROC: 0.605

The Decision Tree improved substantially over KNN, achieving 63.7% accuracy and an AUC-ROC of 0.605. This places it in the marginal-to-acceptable performance range. The model has begun to learn some non-linear decision boundaries in the feature space that are invisible to KNN's distance-based approach.

Confusion Matrix Analysis

	Predicted Non-Potable (0)	Predicted Potable (1)
Actual Non-Potable (0)	362 (True Negatives)	50 (False Positives)
Actual Potable (1)	188 (False Negatives)	56 (True Positives)

The Decision Tree is significantly more conservative than KNN — it predicted potable water far less often (only 56 + 50 = 106 positive predictions out of 656). It correctly identified 362 of 412 non-potable samples (recall = 0.88), but correctly found only 56 of 244 potable samples (recall = 0.23). The model has learned to predict non-potable as a near-default, making it very good at protecting against false positives but very bad at identifying safe water.

Interpretation

The decision tree with max_depth = 5 learned splits that were effective at separating obvious non-potable samples but insufficient to capture the subtle patterns that distinguish potable from non-potable water. The depth constraint prevents overfitting but also limits the model's ability to learn complex feature interactions. Increasing max_depth or using ensemble methods would be expected to improve recall for the potable class.

Random Forest Classifier

Accuracy: 69.4% | AUC-ROC: 0.686

Random Forest is the best-performing model, achieving 69.4% accuracy and an AUC-ROC of 0.686. With an ensemble of 100 trees, the model achieves meaningful improvements over both KNN and Decision Tree across all metrics. The AUC of 0.686 is approaching the commonly used threshold of 0.70 for 'acceptable' model performance.

Confusion Matrix Analysis

	Predicted Non-Potable (0)	Predicted Potable (1)
Actual Non-Potable (0)	366 (True Negatives)	46 (False Positives)
Actual Potable (1)	155 (False Negatives)	89 (True Positives)

Random Forest correctly identified 366 of 412 non-potable samples (recall = 0.89) — comparable to the Decision Tree. Crucially, it also correctly identified 89 of 244 potable samples (recall = 0.36), more than 1.5× the Decision Tree's 56. The False Positive rate (46 samples) is also the lowest of all three models, meaning it is the safest model in terms of incorrectly labelling unsafe water as drinkable.

Interpretation

The ensemble averaging of 100 trees allows Random Forest to capture complex, non-linear feature interactions that no single tree can detect. Each tree learns from a different bootstrap sample of the data and considers a random subset of features at each split, reducing both bias and variance. The result is a model that generalizes substantially better than a single Decision Tree or the KNN approach.

Despite its superiority over the other models tested, the Random Forest's recall for potable water (0.36) remains low. This reflects the fundamental challenge of this dataset: with weak feature-target correlations and class imbalance, even an ensemble of 100 trees struggles to reliably identify the minority potable class.

Key Insights and Recommendations

Insights

The following insights emerged from the EDA, feature analysis phases, to modeling training

Water Potability is a Weak-Signal Problem

When we checked how much each feature (like pH or turbidity) is linked to whether water is potable, the answer was: barely at all. Every single feature had almost zero connection to the target. This tells us that you cannot look at just one measurement and know if water is safe — the relationship is more complex, and only a model that can learn from all features together has a chance of predicting it well.

Class Imbalance Systematically Suppresses Potable-Class Recall

About 61% of samples are labelled non-potable and only 39% are labelled potable. This imbalance means our models saw far more examples of unsafe water during training. As a result, they naturally got better at spotting non-potable water and worse at identifying potable water. This is a known challenge in machine learning called class imbalance, and it directly hurt all three of our models.

Model Complexity Matters

The progression KNN → Decision Tree → Random Forest maps directly onto increasing model complexity and improving performance. This confirms that the water potability classification problem contains meaningful, but non-linear and subtle, patterns in the data that require complex models to detect. Simple models like KNN with Euclidean distance fail to capture these patterns.

All models were better at flagging unsafe water than safe water

Every model we trained was much better at correctly identifying non-potable water than potable water. For example, the best model (Random Forest) correctly identified 89% of unsafe samples but only 36% of safe ones. In plain terms: if you showed the model 100 glasses of safe water, it would correctly say "safe" only about 36 times and wrongly label the other 64 as unsafe. This is the direct effect of the class imbalance mentioned above.

KNN's AUC Below 0.5

The KNN model's AUC-ROC of 0.482 — below the 0.5 baseline of random classification — indicates that KNN working by looking at the nearest neighbours to a new sample in the data doesn’t establish a relationship between the features and the target— as is the case here — "nearness" in the data does not mean anything useful. Two water samples can look very similar by KNN's measure and still belong to different classes. This is why KNN scored below random guessing on the AUC metric (0.482 vs. 0.500 for a coin flip).

Actionable Recommendations

- Perform GridSearchCV or RandomizedSearchCV hyperparameter tuning for the Random Forest (key parameters: n_estimators, max_depth, min_samples_split, max_features). This represents the most promising path to improving the current best model.

- Tune classification thresholds: instead of defaulting to 0.5 probability cutoff, adjust the threshold for the Random Forest to increase recall for potable water, accepting a controlled increase in False Positive rate.

- Look at which features matter most. Random Forest can tell us which features it relied on the most. If we know that, say, pH and turbidity are the most important, water authorities can focus monitoring resources on those measurements.

- Develop a real-time prediction interface (web app or API) that accepts water quality measurements and returns a potability prediction with confidence score.

- Engage water safety domain experts to validate model decisions and ensure predictions align with regulatory frameworks.

Limitations and Future Work

Limitations

- As mentioned earlier, there were far more non-potable samples (1,998) than potable ones (1,278). We did not apply any technique to correct this imbalance before training. As a result, all three models were quietly biased toward predicting non-potable water, which hurt their ability to find safe water samples. This is the biggest weakness of our current approach.

- Several features in the dataset had values far above safe drinking water standards — for example, TDS averaging 22,014 ppm when the safe limit is 500 ppm. This suggests the dataset might include raw or untreated water from rivers, boreholes, or industrial sources, rather than finished drinking water from a treatment plant. If so, the patterns our model learned may not apply well to the kind of water most people actually drink.

- We have no information about where each water sample was collected, what time of year it was, or what type of source it came from. These factors can have a big effect on water quality. For example, river water in the rainy season behaves very differently from borehole water in a dry region. Without this context, our model is essentially working blind to some very important real-world factors.

- We split the data once — 80% for training and 20% for testing. The problem with doing this only once is that the results can vary depending on which samples happened to end up in the test set. If we had split and tested the data five or ten different ways (called cross-validation), we would have a more reliable and trustworthy picture of how well the models actually perform.

- No microbiological parameters (e.g., E. coli, total coliforms) are included, despite being among the most critical determinants of drinking water safety.

Future Work

- Apply stratified SMOTE to the training set to address class imbalance without data leakage.

- Once the model is improved, we would build a simple web application where a user enters nine water quality measurements and gets back a potability prediction — along with a confidence score and an explanation of which features drove the decision. This would make the work practically useful rather than just academic.

- Instead of a single train-test split, we would use 5-fold or 10-fold stratified cross-validation to get more stable and trustworthy performance estimates across the full dataset.

- Collaborate with water utilities and public health agencies to validate predictions against regulatory standards.

Links To LinkedIn Posts

1. Esther Oboh Asamoah - https://www.linkedin.com/posts/esther-oboh-asamoah-80b266290_datascience-machinelearning-firstproject-ugcPost-7448156033384030208-H1Hf?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEaEawsBgkCB7Jmsu6skSW_1Dxxua-SyV9E